

Multiple Choice (10 marks)

1. Statement 1 | If H is a subgroup of a group G and a belongs to G , then $aH = Ha$.
Statement 2 | If H is normal of G and a belongs to G , then $ah = ha$ for all h in H .
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Compute the product in the given ring. $(12)(16)$ in $\mathbb{Z}_{24}$
 - (a) 0
 - (b) 1
 - (c) 4
 - (d) 6
3. For $T: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ where $T(1, 0) = 3$ and $T(0, 1) = -5$, find $T(-3, 2)$.
 - (a) -19
 - (b) -10
 - (c) 19
 - (d) 10
4. Find the characteristic of the ring $\mathbb{Z}_3 \times \mathbb{Z}_3$.
 - (a) 0
 - (b) 3
 - (c) 12
 - (d) 30
5. Statement 1 | Any set of two vectors in $\mathbb{R}^2$ is linearly independent. Statement 2 | If $V = \text{span}(v_1, \dots, v_k)$ and $\{v_1, \dots, v_k\}$ are linearly independent, then $\dim(V) = k$.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The element 4 is a generator for the finite field $\mathbb{Z}_7$.
7. True or False: If $3x + 7y$ is divisible by 11, then $9x + 4y$ must also be divisible by 11 for positive integers x and y .
8. True or False: In the group $G = \{2, 4, 6, 8\}$ under multiplication modulo 10, the identity element is 6.
9. True or False: It is possible to have a nontrivial homomorphism from a finite group into an infinite group.
10. True or False: If g is a group element and $g^n = e$, then the order of g is always equal to n regardless of lower powers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A circle centered at A with a radius of 1 and a circle centered at B with a radius of 4 are externally tangent. A third circle is tangent to the first two and to one of their common external tangents as shown. What is the radius of the third circle? `[asy] draw((-3,0)--(7.5,0)); draw(Circle((-1,1),1),linewidth(0.7)); draw(Circle((3,4),4),linewidth(0.7)); draw(Circle((0.33,1.5),1),linewidth(0.7)); dot((-1,1)); dot((3,4)); draw((-1,1)--(-2,1)); draw((3,4)--(7,4)); label("A",(-1,1),E); label("B",(3,4),W); label("1",(-1.5,1),N); label("4",(5,4),N); [/asy]`
12. Discuss how to determine the maximum area of a triangle formed with one vertex at the center of a circle of radius 1 and the other two vertices on the circle.

End of Paper